

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Numerical

When 9.45 g of ClCH_2COOH is added to 500 mL of water, its freezing point drops by 0.5°C . The dissociation constant of ClCH_2COOH is $x \times 10^{-3}$. The value of x is _____ (Rounded off to the nearest integer) $[K_f(\text{H}_2\text{O}) = 1.86 \text{ K kg mol}^{-1}]$

Q2: 24 Feb (Shift 2) - Numerical

C_6H_6 freezes at 5.5°C . The temperature at which a solution of 10 g of C_4H_{10} in 200 g of C_6H_6 freeze is _____ $^\circ\text{C}$. (The molal freezing point depression constant of C_6H_6 is) 5.12°C/m)

Q3: 25 Feb (Shift 1) - Numerical

1 molal aqueous solution of an electrolyte A_2B_3 is 60% ionised. The boiling point of the solution at 1 atm is _____ K. (Rounded-off to the nearest integer) $[\text{Given } K_b \text{ for } (\text{H}_2\text{O}) = 0.52 \text{ K kg mol}^{-1}]$

Q4: 25 Feb (Shift 2) - Numerical

If a compound AB dissociates to the extent of 75% in an aqueous solution, the molality of the solution which shows a 2.5 K rise in the boiling point of the solution is _____ molal. (Rounded-off to the nearest integer) $[K_b = 0.52 \text{ K kg mol}^{-1}]$

Q5: 26 Feb (Shift 1) - Numerical

224 mL of $\text{SO}_2(\text{g})$ at 298 K and 1 atm is passed through 100 mL of 0.1M NaOH solution. The non-volatile solute produced is dissolved in 36 g of water. The lowering of vapour pressure of solution (assuming the solution is dilute) $(P_{(\text{H}_2\text{O})}^* = 24 \text{ mm of Hg})$ is $x \times 10^{-2}$ mm of Hg, the value of x is (round off to nearest integer)

Q6: 26 Feb (Shift 2) - Numerical

When 12.2 g of benzoic acid is dissolved in 100 g of water, the freezing point of solution was found to be -0.93°C $(K_f(\text{H}_2\text{O}) = 1.86 \text{ K kg mol}^{-1})$. The number (n) of benzoic acid molecules associated (assuming

Questions with Answer Keys

MathonGo

 100% association) is 					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					

Answer Key**Q1 (35)****Q2 (1)****Q3 (375)****Q4 (3)****Q5 (0)****Q6 (2)**

Hints and Solutions

MathonGo

Q1 (35)

Total no. of moles = $c + c\alpha = c(1 + \alpha)$

$$i = \frac{\text{observed}}{\text{calculate}} = \frac{c(1+\alpha)}{c} = (1 + \alpha)$$

M. W. = 94.5

$$\Delta T_f = i \times k_f \times m \quad \Delta T_f = 0.5^\circ\text{C}$$

$$i = 1 + \alpha$$

$$0.5 = (1 + \alpha) \times 1.86 \times \frac{\frac{9.45}{94.5}}{\frac{500}{1000}} \quad m = \frac{\text{mole}}{\text{k.g(Solvent)}}$$

$$k_t = 1.86 \text{ kg/mol}$$

$$(1 + \alpha) = \frac{2.5}{1.86}$$

$$\alpha = \frac{0.64}{1.86} = \frac{32}{93}$$

$$K_a = \frac{C_\alpha^2}{1-\alpha} = \frac{0.2 \times 1024}{93 \times 93 \times \frac{61}{93}}$$

$$K_a = 0.0351 = 35.1 \times 10^{-3}$$

Q2 (1)

$$\Delta T_f = i \times K_f \times m$$

$$= (1) \times 5.12 \times \frac{10/58}{200} \times 1000 \Rightarrow \Delta T_f = \frac{5.12 \times 50}{58} = 4.414$$

$$T_{f(\text{solution})} = T_{K(\text{solvent})} - \Delta T_f$$

$$= 5.5 - 4.414$$

$$= 1.086^\circ\text{C}$$

$$\approx 1.09^\circ\text{C} = 1 \text{ (nearest integer)}$$

Q3 (375)

No. of ions = $2 + 3 = 5$

$$i = 1 + (n - 1) \alpha$$

$$= 1 + (5 - 1) \times 0.6$$

Hints and Solutions

MathonGo

$$= 1 + 4 \times 0.6 = 1 + 2.4 = 3.4$$

$$\Delta T_b = K_b \times m \times i$$

$$= 0.52 \times 1 \times 3.4 = 1.768^\circ\text{C}$$

$$\Delta T_b = (T_b)_{\text{solution}} - [(T_b)_{\text{H}_2\text{O}}]_{\text{Solution}}$$

$$1.768 = (T_b)_{\text{solution}} - 100$$

$$(T_b)_{\text{solution}} = 101.768^\circ\text{C}$$

$$= 375\text{ K}$$

Q4 (3)



$$1 - \alpha \quad \alpha \quad \alpha$$

$$\alpha = 3/4$$

$$N = 2$$

$$i = [1 + (2 - 1)\alpha]$$

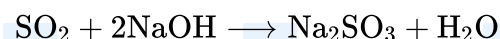
$$2.5 = [1 + (2 - 1)3/4] \times 0.52 \times m$$

$$m = \frac{2.5}{0.52 \times 7/4} = \frac{10}{3.64} = 2.747$$

$$m = 2.747 \simeq 3 \text{ mol/kg}$$

Q5 (0)

The balanced equation is



$$\text{moles of NaOH} = \text{molarity} \times \text{volume (in litre)}$$

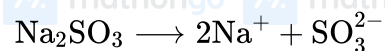
$$= 0.1 \times 0.1$$

$$= 0.01 \text{ moles}$$

Here NaOH is limiting Reagent $2 \text{ mole NaOH} \longrightarrow 1 \text{ mole Na}_2\text{SO}_3$

$$0.01 \text{ mole NaOH} \longrightarrow \frac{1}{2} \times 0.01 \text{ mole Na}_2\text{SO}_3$$

$$\text{Moles of Na}_2\text{SO}_3 \longrightarrow 0.005 \text{ mole}$$



$$i = 3$$

$$\text{Moles of H}_2\text{O} = \frac{36}{18} = 2 \text{ moles}$$

According to RLVP -

Hints and Solutions

MathonGo

$$\frac{P_A^0 - P_A}{P_A^0} = i X_B$$

$$\frac{P_A^0 - P_A}{P_A^0} = \frac{i n_A}{i n_A + n_B} (i n_A \simeq 0)$$

$$n_B \ll n_A$$

$$\{n_A + n_B \simeq n_A\}$$

$$\frac{P_A^0 - P_A}{P_A^0} = i \times \frac{n_B}{n_A}$$

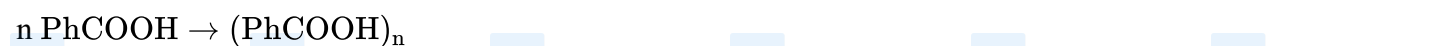
$$\frac{2H - P_A}{2H} = 3 \times \frac{0.005}{2}$$

$$\Rightarrow 2H - P_A = 0.18$$

Lowering in pressure = 0.18 mm of Hg lowering in pressure = 18×10^{-12} mm of Hg

$$X = 18$$

Q6 (2)



$$N = \frac{1}{X} = i \{ \text{As } \alpha = 1 \}$$

$$\Delta T_f = i \times k_f \times m$$

$$0.93 = \frac{1}{n} \times 1.86 \times \frac{12.2 \times 1000}{122 \times 100}$$

$$n = 2$$